13. Domain: Software Technologies including BOSS

Sr. No.	Parameters	Brief details
1.	**Brief details of Technology Developed**	C-DAC Revival Disaster Recovery product family consists of a variety of data replication and backup solutions to provide end to end disaster recovery for enterprise business applications. C-DAC Revival products can be used to manage planned and unplanned outage, enabling 24x7 data availability for business. It provides long term business continuity, continuous data availability and customer satisfaction ensuring near zero RPO and negligible RTO for the business-critical data.
	Name of the Project under which technology is developed	e-Sangam and Multi-Site Disaster Recovery as a Service on Cloud
	Funding Agency	MeitY
	Technology Readiness Level (TRL)	9
	Brief Write-up on the Technology Developed	**Revival Block Replicator** C-DAC Revival Block Replicator is an end-to-end DR solution, best suited for database intensive e-Gov and Business applications. C-DAC Revival makes use of synchronous and semi-synchronous replication to achieve optimal business continuity with zero Recovery Point Objective (RPO) and negligible Recovery Time Objective (RTO). It uses iSCSI over TCP/IP to provide block replication at remote sites. It is interoperable with legacy hardware systems including SAN boxes and works very well with database-based applications. Revival Block Replicator Products are as follows. **Revival Sync:** Real-time synchronous block replication over IP network from C-DAC Revival DC to C-DAC Revival DR located within 50 km to provide zero recovery point objective. **Revival Semi-sync/Async:** Real-time semi-synchronous block replication over IP network from C-DAC Revival DC to C-DAC Revival DR over WAN distances and provides negligible recovery point objective bounded by network delay. **Revival Optimal DR:** Combination of synchronous and semi-synchronous block replication over IP network with three tier architectures to provide zero recovery point objective over WAN. **Revival Flat-File Backup** Revival flat-file backup is a platform independent file level replication/backup solution. The lightweight Revival agent captures the recent file-based transactions from the source and transfers it periodically to the destination. The data integrity check is maintained between source and destination to ensure the data consistency. This solution is best suited for various file level data formats. **Revival Active-Active Replicator** Revival active-active replicator is best suited for database-based applications having active load at both primary and secondary sites. The near distance

Sr. No.	Parameters	Brief details
		active database data is bidirectionally replicated between primary and secondary sites using asynchronous mechanisms. Revival active-active replicator is also suitable for cross platform database applications. The consistency check at both ends ensures data integrity. **Revival DRM: Software Web Solution for DR Management and Monitoring** All the above-mentioned Revival replication solutions are managed and monitored using a centralized DR management and monitoring web service application called C-DAC DRM. C-DAC DRM is designed to manage DR related drill activities such as Normal copy, Reverse normal copy, Switchover, Switchback, Failover, Failback, Replication status, Jobs, RPO and RTO status etc. These activities are fully automated through C-DAC DRM. One can monitor the entire DR setup for their various DR deployments through centralized DRM. This reduces manual intervention which is often prone to human errors and helps in effortless monitoring of the entire DR setup. DRM provides report generation of the activities carried out and alert/notification for critical events via email.
	Details of Contact Person alongwith Mobile No. and e-mail ID	Dr. Shreya Bokare 9082165625 shreya@cdac.in
2.	**Name of the Project under which technology is developed**	e-Hastakshar: C-DAC's On-line Digital Signing Service
	Funding Agency	Ministry of Electronics and Information Technology (MeitY)
	Technology Readiness Level (TRL)	9
	Brief Write-up on the Technology Developed	As a flagship initiative within the Government's Digital India program, C-DAC has introduced e-Hastakshar, a cutting-edge eSign service that allows citizens to digitally sign documents online in real-time, providing a legally acceptable form and convenient alternative to physical signatures. Over the past year, C-DAC integrated this service with various departments, ministries, and agencies at the Central and State Government levels, as well as Union Territories. C-DAC utilizes service of Unique Identification Authority of India (UIDAI) for on-line authentication and Aadhaar eKYC service. **e-Hastakshar service supports Online Aadhaar Authentication Modes - One-time password (OTP), TOTP, Fingerprint, IRIS, Face (Mobile Apps only)** based modes of authentication for leveraging eKYC service of UIDAI.
	Details of Contact Person, C-DAC Centre along with Mobile No. and e-mail ID	Jahnavi Bodhankar, C-DAC Pune 9860263344, jahnavib@cdac.in

Sr. No.	Parameters	Brief details
3.	**Brief details of Technology Developed**	Indigenous hyper-converged cloud platform providing compute, storage, and networking using open-source technologies.
	Name of the Project under which technology is developed	C-DAC Virtue HCI
	Funding Agency	Internal Funding
	Technology Readiness Level (TRL)	TRL -9
	Brief Write-up on the Technology Developed	The Virtue HCI platform integrates compute, storage, and networking into a single software-defined stack. It supports VM management, distributed storage, role-based access control, and centralized monitoring. The platform is designed for government and enterprise data centers requiring a secure, indigenous cloud solution.
	Details of Contact Person, C-DAC Centre along with Mobile No. and e-mail ID	Mr. Sakthi Saravanakumar P., C-DAC Chennai 9840736578 sakthis@cdac.in
4.	**Brief details of Technology Developed**	MeghSikshak is an indigenously developed e-learning platform designed for offering online courses. It serves as a comprehensive solution for modern learning and development needs, catering to various aspects of online education such as skill upgrading, capacity building, training, campaigns, credential programs, and knowledge enhancement. It is a versatile tool empowering educators, trainers, and organizations with innovative features that enhance the learning experience.
	Name of the Project under which technology is developed	NA
	Funding Agency	NA
	Technology Readiness Level (TRL)	TRL-8
	Brief Write-up on the Technology Developed	MeghSikshak provides the essential Learning Management System (LMS) services such as Sharable Content Object Reference Model (SCORM) based Content Delivery, Question & Test Interoperability (QTI) based online Assessment, Assignment, Learning Analytics, Discussion Forum, Certificate Generation and other communication & collaboration services. • Fully Customizable • Secure and Scalable • Microservices based architecture • Can be easily Integrated with third party services/APIs • Cloud Compatible deployment • Multilingual Support • Standards Compliance • Responsive UI Mobile Application
	Details of Contact Person alongwith Mobile No. and e-mail ID	Mr Ramu Parupalli, C-DAC Hyderabad ramup@cdac.in +91 9966180312

Sr. No.	Parameters	Brief details
5.	**Brief details of Technology Developed**	Vikaspedia is an AI enabled Multilingual, Multimodal decentralised Content Management framework that facilitates collaborative knowledge sharing among diverse stakeholders. (Wiki model)
	Name of the Project under which technology is developed	Vikaspedia
	Funding Agency	Ministry of Electronics and Information Technology (MeitY)
	Technology Readiness Level (TRL)	Level-8
	Brief Write-up on the Technology Developed	The AI enabled Vikaspedia collaborative knowledge management framework is best suited for offering citizen-centric multilingual knowledge sharing platforms and managing Institutional knowledge repositories. The key features include • Microservices based architecture, Secure & scalable, Integration with third party services/APIs, Single sign on (SSO) facility, Responsive UI • Supports all 22 constitutionally recognized languages of India, besides English. Supports Internationalized Domain Names. • Supports all major content forms – Text, images, Multimedia • Provides for an AI enabled and role-based framework for content authoring, review, discovery and personalization. • Enables user interaction through features like notifications, feedback, page rating and commenting, connecting with social media, etc. • Compliant to web and accessibility standards
	Details of Contact Person alongwith Mobile No. and e-mail ID	Mrs Vijayalakshmi B, C-DAC Hyderabad vijayab@cdac.in +91 9490118137
6.	**Brief details of Technology Developed**	Education and Training Software Suite
	Name of the Project under which technology is developed	Deployed Post Graduate Diploma Courses at Advanced Computing Training School (ACTS)
	Funding Agency	Ministry of External Affairs (MEA)
	Technology Readiness Level (TRL)	8
	Brief Write-up on the Technology Developed	A suite of education and training products and services for entrance exam registration, exam conduction, lab assessments, feedback assessment, analytics and visualization dashboards, placement management, payment management and administration, identity access management, and learning management. • Key Features/ Unique Selling Point of the Product/ Solution: ○ An end-to-end integrated web-based software solution taking care of all educational activities right from student selection, the learning journey until industry placement. It's a one-stop solution for all ACTS department with service-level implementations and open-APIs providing flexibility to various stakeholders for integration and delivery.

Sr. No.	Parameters	Brief details
		○ A solution to handle the registration process of candidates applying for C-DAC's PG Diploma courses. ○ A Learning Management System (LMS) that encompasses training, education, just-in-time information and communication, computer-based learning, virtual classrooms and digital collaborations. ○ A platform for hosting courses and self-paced learning environment. ○ An application which can be used for getting feedback about faculty, infrastructure and other services offered by CDAC from students studying at C-DAC centres and also at the various Authorized Training Centres (ATC) affiliated to C-DAC for different training programs offered by CDAC ACTS, which makes feedback collection and analysis easy. ○ An application for automation of the examination process. It assists in preparing question papers, scheduling exams, monitoring exams and generating results. ○ A system automates the manual processes of carrying out placement and recruitment. ○ A user-centric web platform with data analytics and reporting capabilities for several departments and user groups. Analytics and Visualization dashboards can be aesthetically customized for Legends, Colors, Sizes etc. Filtering/ Sorting techniques can be enabled to analyze data as per requirement and interest. Alerts can be configured in a way that if an anomaly occurs, they are activated and these events can be reported to the monitoring team through email.
	Details of Contact Person, C-DAC Centre along with Mobile No. and e-mail ID	Parimal Wagh, C-DAC Pune, 9422987975 parimalw@cdac.in
7.	**Brief details of Technology Developed**	BOSS is an indigenous operating system has been developed for the nation. BOSS was initially released in 2007 and has evolved over the years through multiple upgrades and enhancements. BOSS is available in several variants, including Desktop, Secure Enterprise, Server, and an operating system for High Performance Computing (HPC). The BOSS Enterprise version has been developed to meet strategic requirements and is actively deployed within the Indian Army, Navy, and other defense organizations.
	Name of the Project under which technology is developed	Bharat Operating System Solutions
	Funding Agency	MeitY
	Technology Readiness Level (TRL)	TRL-5
	Brief Write-up on the Technology Developed	BOSS (Bharat Operating System Solutions) is an indigenous Linux-based operating system developed to provide a secure, reliable, and self-reliant computing platform for national requirements, with tailored variants supporting desktop, enterprise, server, and high-performance computing needs, including strategic defense deployments.

Sr. No.	Parameters	Brief details
	Details of Contact Personalongwith Mobile No. and e-mail ID	Vidhiyalakshmi A, C-DAC Chennai vidhiyalakshmia@cdac.in 9442510095